AI

BIOMETRIC ATTENDANCE SYSTEM

A Real Time Face Recognition Solution for Attendance

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Final Internship Project [FLEXISAF]

PROBLEM STATEMENT

- Manual Fraud (Buddy Punching): Traditional systems allow employees or students to clock in for one another using shared ID cards, passwords, or manual signatures.
- Inaccurate Labor Costs: Companies lose significant revenue paying for "ghost hours" where an employee is logged in but not physically present.
- Security Vulnerabilities: Lack of biometric verification means unauthorized individuals can easily bypass non-visual security checks.
- Time Inefficiency: Manual verification is slow and creates bottlenecks during peak entry hours (e.g., 8:00 AM).

THE SOLUTION

A THREE-PILLAR AI ECOSYSTEM

TO ELIMINATE BUDDY PUNCHING AND MANUAL ERRORS, THE SYSTEM IS BUILT ON THREE INTEGRATED MODULES

1. AI SCANNER (THE ENGINE)

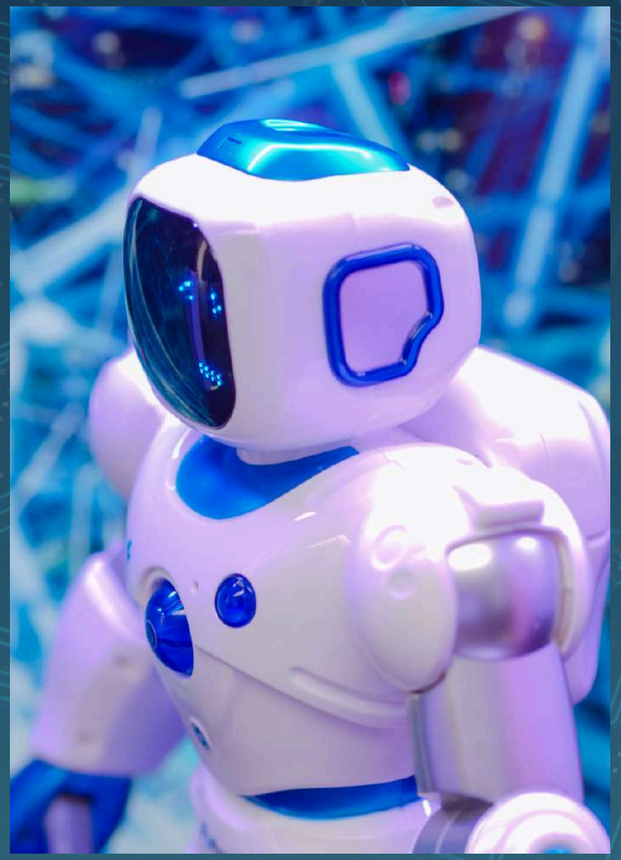
- Real-time Detection: High-speed scanning using HOG-based geometry to find faces in under 0.1 seconds.
- Instant Identification: Comparison of live 128-D biometric signatures against the authorized "Student DNA" database.

2. SECURITY HUB (THE CONTROLLER)

- Automated Protocols: Integrated "Auto-Shutdown" feature that triggers if an unauthorized individual persists in the frame for >5 seconds.
- Privacy-First Design: All biometric math is processed locally on the "Edge" (the laptop) to ensure data never leaks to the cloud.

3. ANALYTICS DASHBOARD (THE MANAGER)

- Web-Based Portal: A professional UI built with Streamlit for administrators to monitor logs.
- Intelligent Tracking: Automated Absentee List and Attendance Rate calculations using Set Mathematics.



TECH STACK [THE TOOLS]

- Operating System: Ubuntu 24.04 (Linux) - Chosen for high-performance AI math.
- Language: Python 3.12
- AI Libraries: Dlib (Math engine), Face_Recognition (Wrappers), OpenCV (Video).
- Web Framework: Streamlit (Frontend Dashboard).

THE SCIENCE

DETECTION (HOG)

- **Finding Shapes**: The AI ignores skin color and looks at the unique geometry of the face, histogram of oriented gradient(HOG).
- **Mapping Features**: It identifies 68 tiny "landmarks" (like a connect-the-dots game) to ensure it's a real person.
- **The Digital Signature**: It converts your face into a 128-number secret code.
- **The Security**: Unless those 128 numbers match perfectly, the door stays closed.

THE AI SCANNER

HIGH-SPEED, HIGH ACCURACY ENTRY

Real time Recognition: No waiting. It scans and verifies in a blink.

Smart Recognition: It can tell the difference between you and a photo or even a twin.

Visual Feedback: Clear Green/Red indicators so users know they are "Access Granted."

THE SECURITY HUB

A SYSTEM THAT PROTECTS ITSELF

1.

THE UNKNOWN ALERT

If a stranger stands in front of the camera, the system detects a "Security Anomaly."

2.

AUTO LOCKDOWN

After 5 seconds of unauthorized presence, the system automatically shuts down and triggers a voice warning.

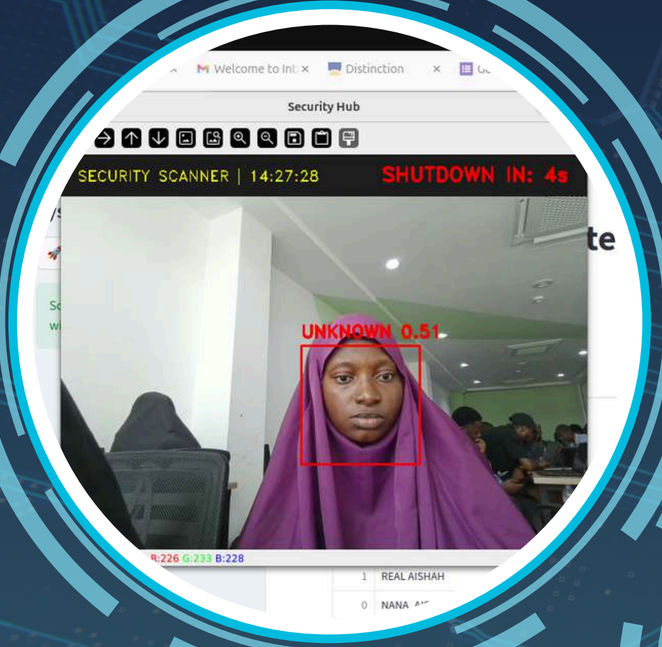
THE MANAGEMENT DASHBOARD

1. **VISUAL REPORTS**
   See beautiful charts showing who is present and who is late.

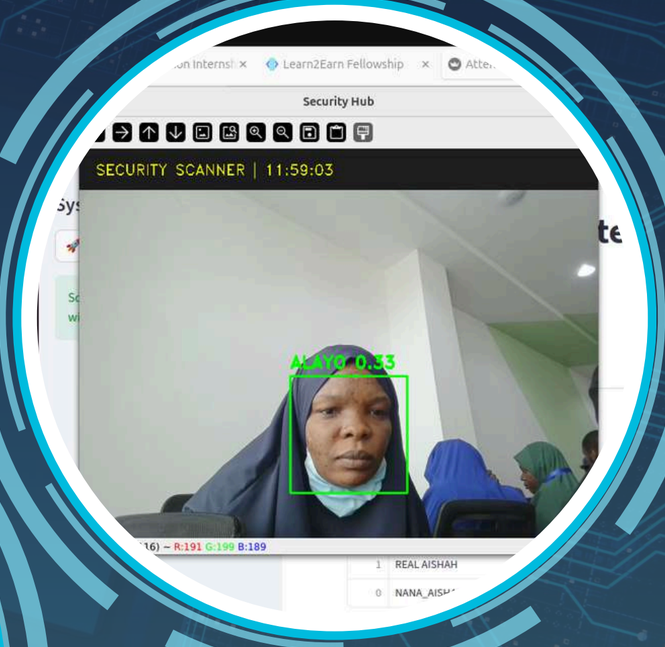
2. **THE RED ALERT LIST**
   Instantly see a list of Absentees, no more manual searching

3. **DIGITAL LOGS**
   A permanent, unchangeable record of every entry and exit.

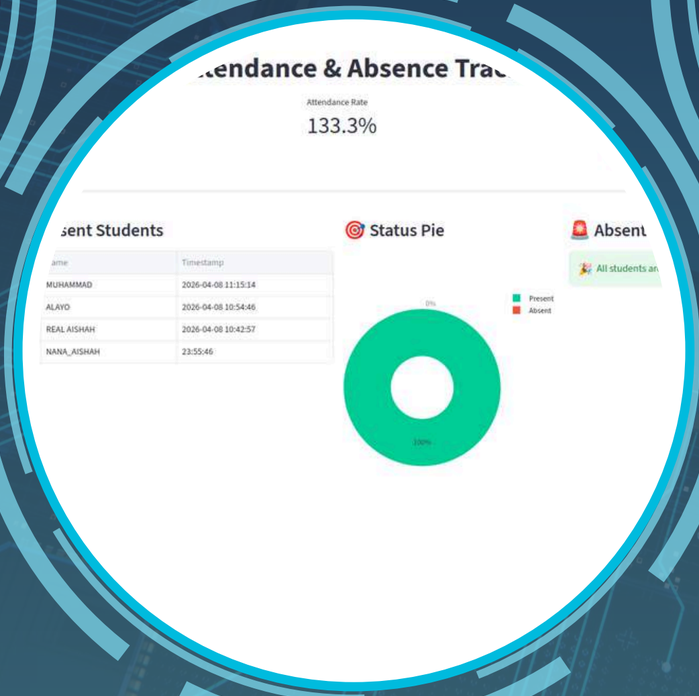
VISUAL PROOF (SCREENSHOTS)



Screenshot of the red box and countdown for a stranger



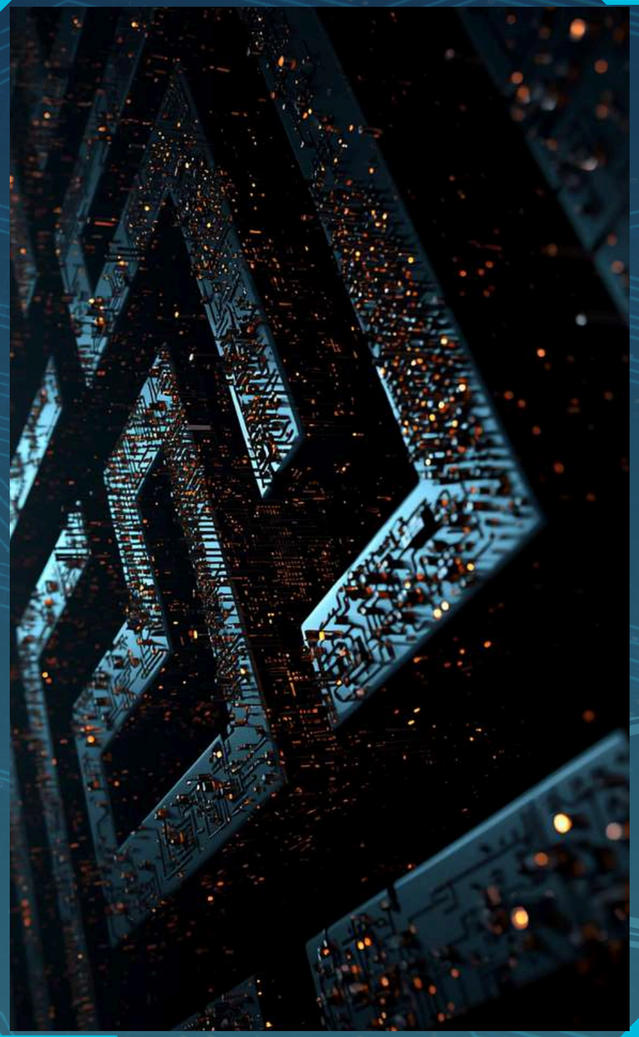
Screenshot of the green box recognizing student/staff



Dashboard with the pie chart and the absentee list.

OVERCOMING HURDLES

- **Connecting the Brain**: Initially, the computer couldn't find the AI's memory files. I had to manually show the software exactly where to find its "knowledge" so it could start recognizing faces.
- **Teaching Strictness**: At first, the AI was too friendly it might have mistaken a stranger for a student. I adjusted the settings to make the AI much stricter, so it only grants access when it is 100% sure.
- **Camera Permission**: Sometimes the camera wouldn't turn on because the computer was protecting it. I fixed the permissions so the camera and the software can talk to each other perfectly.



FINAL THOUGHTS

- The Biometric Hub is more than just code; it's a reliable team member.
- It kills "Buddy Punching" by ensuring only the right people get in.
- It kills Paperwork by doing the counting and sorting for you.
- It provides Peace of Mind by watching the door so you don't have to.

THANK YOU

FOR LISTENING